13. Coagulation or precipitation

The flocculation and settling down of the sol particles is called coagulation.

Various method of coagulation are given below:

(i) Addition of electrolytes

(ii) Electrophoresis

(iii) Mixing oppositively charged sols.

(iv) Boiling

(1) Addition of electrolytes

A negative ion causes the precipitation of positively charged sol and vice versa.

When the valency of ion is high, the precipitation power is increased. For example, the precipitation power of some cations and anions varies in the following order

$$Al^{3+} > Ba^{2+} > Na^{+} \text{ , Similarly } [Fe(CN)_6]^{3-} > SO_4^{2-} > Cl^{-}$$

The precipitation power of electrolyte is determined by finding the minimum concentration (millimoles/lit) required to cause precipitation of a sol in 2hours. This value is called flocculation value. The smaller the flocculation value greater will be precipitation.

(ii) Electrophoresis:

In the electrophoresis, charged particles migrate to the electrode of opposite sign. It is due to neutralization of the charge of the colloids. The particles are discharged and so they get precipitated.

(iii) By mixing two oppositively charged sols

When colloidal sols with opposite charges are mixed mutual coagulation takes place. It is due to migration of ions from the surface of the particles.

(iv) By boiling

When boiled due to increased collisions, the sol particles combine and settle down.

14. Protective action

Generally, lyophobic sols are precipitated readily even with small amount of electrolytes. But they are stabilised by addition of a small amount of lyophillic colloid.

A small amount of gelatine sol is added to gold sol to protect the gold sol.

Colloid	Gold number
Gelatin	0.005-0.01
Egg albumin	0.08-0.10
Gum Arabic	0.1-0.15
Potato starch	25

Zsigmondy introduced the term 'gold number' as a measure of protecting power of a colloid. Gold number is defined as the number of milligrams of hydrophilic colloid that will just prevent the precipitation of 10ml of gold sol on the addition of 1ml of 10% NaCl solution. Smaller the gold number greater the protective power.

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